SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5 **Duration : 1 Hour**
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I __J.Jeshwanth(1921260264)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

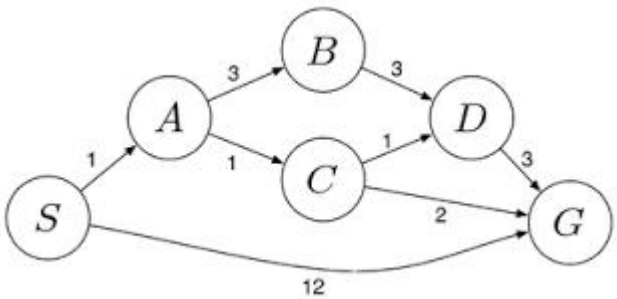
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Artificial Intelligence – Problem-Solving Agent Answers

1. Water Jug Problem (4-gallon and 3-gallon jugs)

Goal: Obtain exactly 2 gallons of water in the 4-gallon jug.

Assumptions: The jugs can be filled, emptied, and poured into each other. No measuring devices are available.

Solution steps:

Step	Action	4-gallon jug	3-gallon jug
1	Fill the 3-gallon jug	0	3
2	Pour into the 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Pour into the 4-gallon jug until it is full	4	2
5	Empty the 4-gallon jug	0	2
6	Pour the remaining water into the 4-gallon jug	2	0

Result: Exactly 2 gallons of water are in the 4-gallon jug.

2. Mars Rover as an Intelligent Agent

i. What are the percepts of this agent?

Percepts are the information received from the environment through sensors.

Images of the Martian surface.

Soil and rock composition.

Temperature, pressure, and radiation.

Distance to obstacles and terrain conditions.

Rover position, battery level, and equipment status.

ii. Characterize the operating environment

The Mars Rover operates in an environment that is:

Partially observable: It cannot observe everything around it.

Stochastic: Outcomes can be uncertain due to terrain and equipment conditions.

Sequential: Current actions affect future actions.

Dynamic: Environmental conditions may change.

Continuous: Position, time, and sensor measurements vary continuously.

Single-agent: The rover performs its mission independently of other intelligent agents.

iii. What actions can the agent take?

Move forward, backward, and turn.

Capture images and scan terrain.

Collect soil and rock samples.

Analyze samples using scientific instruments.

Avoid obstacles and select safe routes.

Transmit collected data to Earth.

Recharge or conserve power where possible.

iv. How can the performance be evaluated?

Performance can be measured by:

Number and quality of samples collected and analyzed.

Scientific discoveries and useful data transmitted.

Distance explored and mission objectives completed.

Safety and avoidance of equipment damage.

Efficient use of energy and mission time.

v. Most suitable agent architecture

Model-based, goal-based agent with utility-based decision-making.

It maintains an internal model of the environment, selects actions to achieve scientific goals, and evaluates trade-offs between safety, scientific value, time, and energy. This is more suitable than a simple reflex agent because the rover must plan under uncertainty.

3. Eight-Queens Problem: Problem Formulation and Search Space

Problem: Place 8 queens on an (8 \times 8) chessboard so that no two queens attack each other.

Problem components

1.

Initial state: An empty chessboard.

State space: All possible arrangements of queens on the chessboard.

Initial action: Place a queen in the first column.

Actions: Place a queen in the next column in a row that does not conflict with previously placed queens.

Transition model: Adds a queen to the selected square.

Goal test: All 8 queens are placed, with no two queens sharing a row, column, or diagonal.

Path cost: Usually one unit per queen placed.

Search strategy: Backtracking

Start with an empty board.

Place a queen in a safe square in the first column.

Move to the next column and place another queen in a safe square.

If no safe square exists, remove the previous queen and try another position.

Continue until all 8 queens are placed.

Example solution: Represent each column by the row containing its queen.

[[1,5,8,6,3,7,2,4]]

This is a valid arrangement in which no two queens attack each other.

Conclusion: Backtracking systematically explores possible configurations and rejects invalid states early, reducing unnecessary search.

4. OLA Cab Booking Using a Problem-Solving Agent

Type of agent

A utility-based problem-solving agent is most suitable. It evaluates available cab options based on cost, travel time, comfort, availability, and customer preferences, then selects the most suitable option.

Pseudocode

```
START

INPUT source, destination
INPUT customer_preferences

cab_types = [Mini, Micro, Sedan, Shared, Prime]
available_cabs = FIND_AVAILABLE_CABS(source, destination)

best_cab = NONE
best_utility = -INFINITY

FOR each cab IN available_cabs:
    fare = ESTIMATE_FARE(cab, source, destination)
    time = ESTIMATE_TRAVEL_TIME(cab, destination)
    comfort = GET_COMFORT_SCORE(cab)
    utility = CALCULATE_UTILITY(
        fare, time, comfort, customer_preferences
    )

    IF utility > best_utility:
        best_utility = utility
        best_cab = cab

DISPLAY best_cab, fare, time
```

```
IF customer_confirms:
   BOOK_CAB(best_cab)
   DISPLAY "Cab booked successfully"

END
```

Conclusion: The utility-based agent compares alternatives and recommends a cab that best matches the customer's requirements.

5. Uniform Cost Search (UCS) – Delivery Network

Objective: Find the minimum-cost path from starting node (S) to goal node (G).

The weighted graph provided in the question contains these edges:

(S \rightarrow A = 1)

(S \rightarrow G = 12)

(A \rightarrow B = 3)

(A \rightarrow C = 1)

(B \rightarrow D = 3)

(C \rightarrow D = 1)

(C \rightarrow G = 2)

(D \rightarrow G = 3)

UCS exploration

Uniform Cost Search always expands the node with the lowest cumulative path cost (g(n)).

Step	Node expanded	Path cost	Frontier candidates
1	S	0	A(1), G(12)
2	A	1	C(2), B(4), G(12)
3	C	2	D(3), G(4), B(4)
4	D	3	G(4), B(4), G(6)
5	G	4	Goal reached

Note: Equal-cost nodes can be expanded in either order. Duplicate paths to a node are handled by retaining the least-cost path.

Optimal path

[S \rightarrow A \rightarrow C \rightarrow G]

Total cost:

[1+1+2=\boxed{4}]

Final result: UCS finds the least-cost route (S \rightarrow A \rightarrow C \rightarrow G), with a total delivery cost of 4 units, which is cheaper than the direct route costing 12 units.